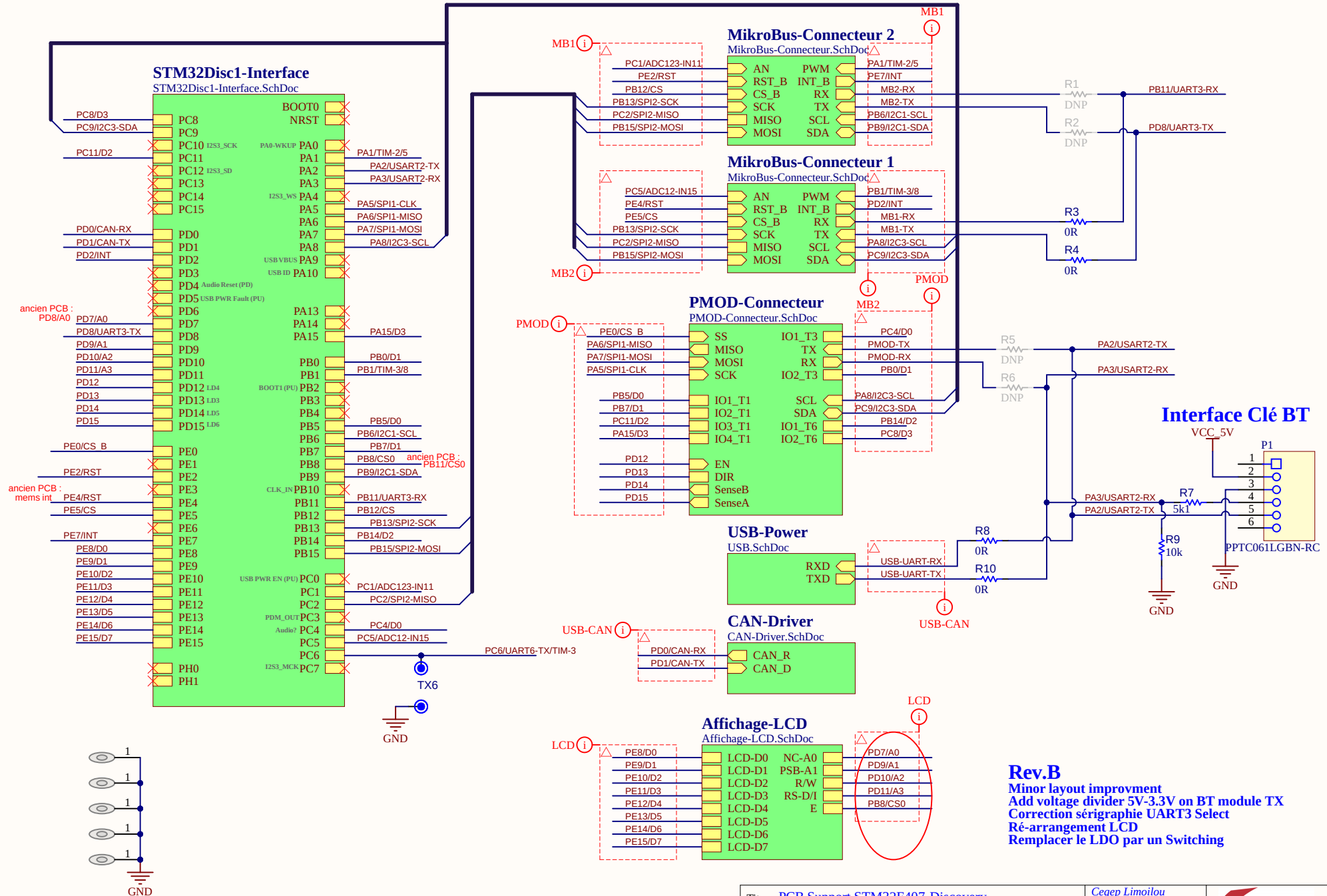


PCB Support STM32F407-Discovery



Titre : PCB Support STM32F407-Discovery

Conception :
CEGEP Limoilou

Date : 2026-02-27

Fichier : PCB-Support-STM32F407.SchDoc

Approbation :
kc

Heure : 13:54:46

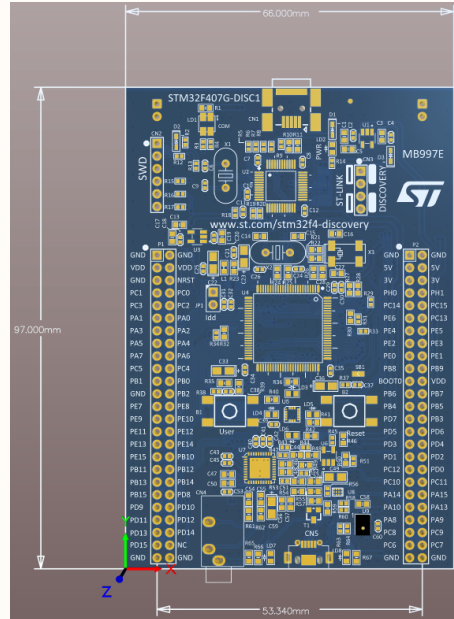
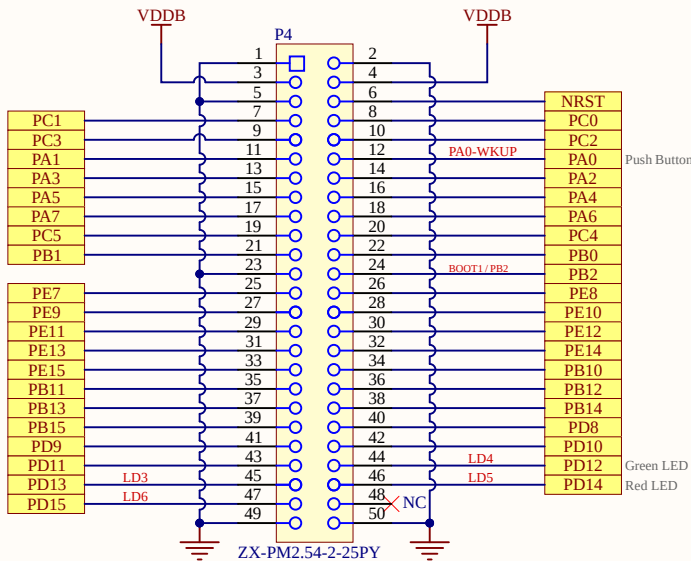
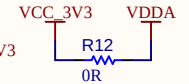
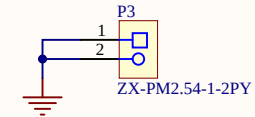
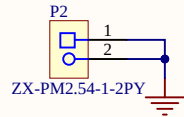
Révision : Rev.B

Feuille 1 de 10

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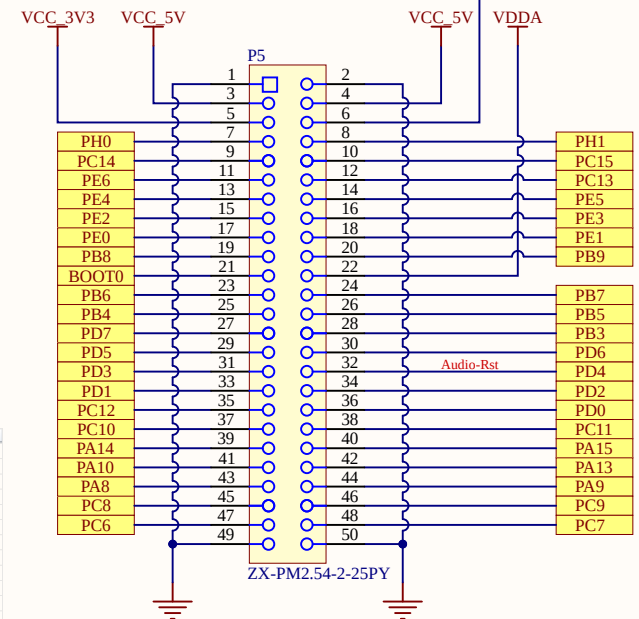


STM32 DISCOVERY 407



Position	Name	Type	Signal	Label
23	PA0-WKUP	I/O	GPIO_EXTIO	B1 [Blue PushButton]
69	PA10	I/O	USB_OTG_FS_ID	OTG_FS_ID
70	PA11	I/O	USB_OTG_FS_DM	OTG_FS_DM
71	PA12	I/O	USB_OTG_FS_DP	OTG_FS_DP
72	PA13	I/O	SYS_JTMS-SWDIO	SWDIO
76	PA14	I/O	SYS_JTCK-SWCLK	SWCLK
29	PA4	I/O	I2S3_WS	I2S3_WS[CS43L22_LRCK]
30	PA5	I/O	SP11_SCK	SP11_SCK[US302DL_SCLK]
31	PA6	I/O	SP11_MISO	SP11_MISO[US302DL_SDO]
32	PA7	I/O	SP11_MOSI	SP11_MOSI[US302DL_SDASD/SDO]
68	PA9	I/O	USB_OTG_FS_VBUS	VBUS_FS
47	PB10	I/O	I2S2_CLK	CLK_IN[MP450T02_CLK]
37	PB2	Input	GPIO_Input	BOOT1
89	PB3	I/O	SYS_JTDO-SWO	SWO
92	PB6	I/O	I2C1_SCL	Audio_SCL[CS43L22_SCL]
96	PB9	I/O	I2C1_SDA	Audio_SDA[CS43L22_SDA]
15	PD0	Output	GPIO_Output	OTG_FS_PowerSwitchOn
78	PD10	I/O	I2S3_CLK	I2S3_SCK[CS43L22_SCLK]
80	PD12	I/O	I2S3_SD	I2S3_SD[CS43L22_SDM]
8	PC14-OSC32_IN	I/O	RCC_OSC32_IN	PC14-OSC32_IN
9	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	PC15-OSC32_OUT
18	PC3	I/O	I2S2_SD	PDMA_OUT[MP450T02_DOUT]
64	PC7	I/O	I2S3_MCK	I2S3_MCK[CS43L22_MCLK]
59	PD12	Output	GPIO_Output	LD1 [Green Led]
60	PD13	Output	GPIO_Output	LD3 [Orange Led]
61	PD14	Output	GPIO_Output	LD5 [Red Led]
62	PD15	Output	GPIO_Output	LD6 [Blue Led]
85	PD4	Output	GPIO_Output	Audio_RST[CS43L22_RESET]
86	PD5	Input	GPIO_Input	OTG_FS_OverCurrent
98	PE1	I/O	GPIO_EXTI1	MEMS_INT2[US302DL_INT2]
2	PE3	Output	GPIO_Output	CS_I2C/SP1[US302DL_CS_I2C/SP1]
12	PH0-OSC_IN	I/O	RCC_OSC_IN	PH0-OSC_IN
13	PH1-OSC_OUT	I/O	RCC_OSC_OUT	PH1-OSC_OUT

The JTAG pins are in input PU/PD after reset:
 PA15: JTDI in PU
 PA14: JTCK in PD
 PA13: JTMS in PU
 PB4: NJTRST in PU



Titre : STM32 DISCOVERY 407

Conception : CEGEP Limoilou

Approbation : kc

Révision : Rev.B

Date : 2026-02-27

Heure : 13:54:46

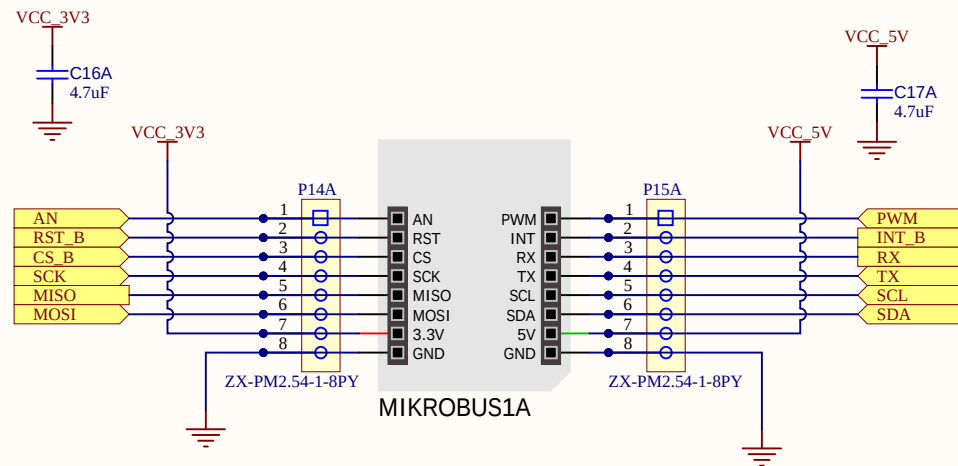
Feuille 2 de 10

Fichier : STM32Disc1-Interface.SchDoc

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Mikro Bus - Connecteur



Titre : Mikro Bus - Connecteur

Conception :
CEGEP Limoilou

Date : 2026-02-27

Fichier : MikroBus-Connecteur.SchDoc

Approbation :
kc

Heure : 13:54:47

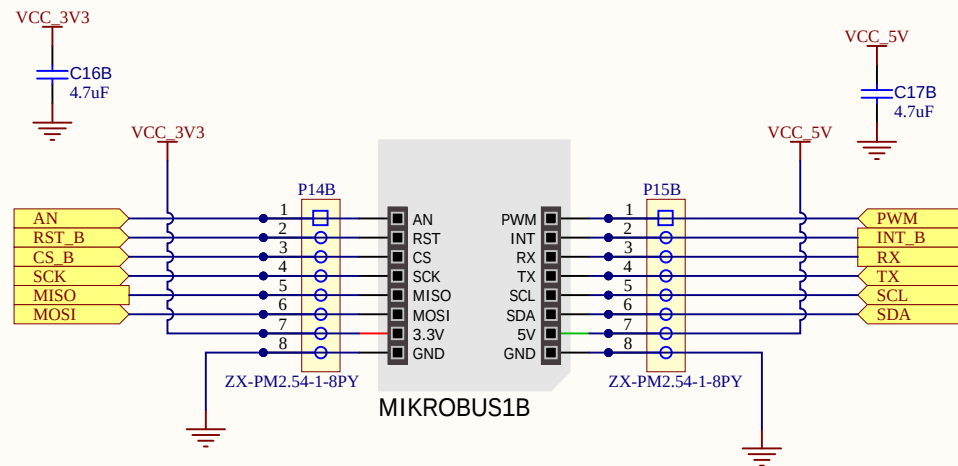
Révision : Rev.B

Feuille 3 de 10

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Mikro Bus - Connecteur



Titre : Mikro Bus - Connecteur

Conception :
CEGEP Limoilou

Date : 2026-02-27

Fichier : MikroBus-Connecteur.SchDoc

Approbation :
kc

Heure : 13:54:47

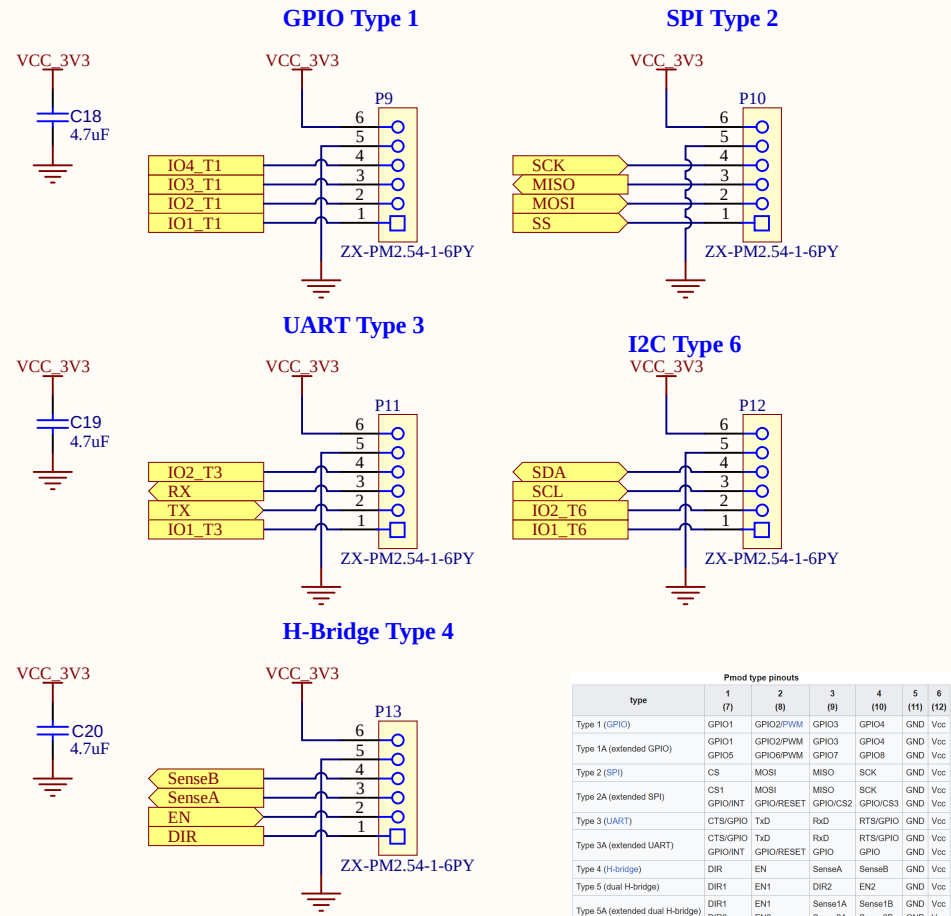
Révision : Rev.B

Feuille 3 de 10

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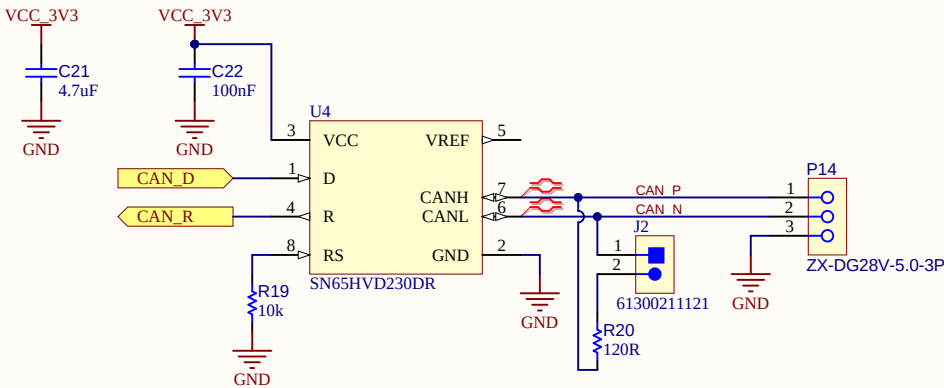


Connecteurs - PMOD Interface



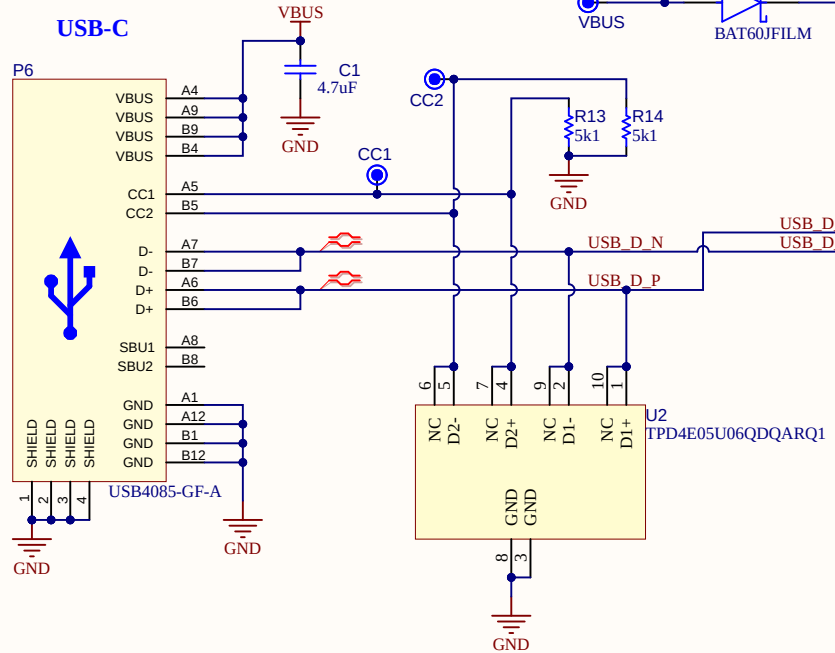
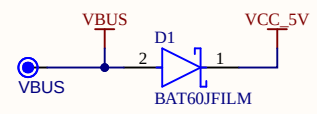
Pmod type pinouts						
type	1 (7)	2 (8)	3 (9)	4 (10)	5 (11)	6 (12)
Type 1 (GPIO)	GPIO1	GPIO2/PWM	GPIO3	GPIO4	GND	Vcc
Type 1A (extended GPIO)	GPIO1	GPIO2/PWM	GPIO3	GPIO4	GND	Vcc
Type 2 (SPI)	CS	MOSI	MISO	SCK	GND	Vcc
Type 2A (extended SPI)	CS1	MOSI	MISO	SCK	GND	Vcc
Type 3 (UART)	CTS/GPIO	TxD	RxD	RTS/GPIO	GND	Vcc
Type 3A (extended UART)	CTS/GPIO	TxD	RxD	RTS/GPIO	GND	Vcc
Type 4 (H-bridge)	DIR	EN	SenseA	SenseB	GND	Vcc
Type 5 (dual H-bridge)	DIR1	EN1	DIR2	EN2	GND	Vcc
Type 5A (extended dual H-bridge)	DIR1	EN1	Sense1A	Sense1B	GND	Vcc
Type 6 (I2C)	nc/INT	nc/RESET	SCL	SDA	GND	Vcc
Type 6A (extended I2C)	nc/INT	nc/RESET	SCL	SDA	GND	Vcc
Type 7 (I2S)	LRCLK	DACdata	ADCLdata	BCLK	GND	Vcc
	GPIO	GPIO	MCLK	GPIO	GND	Vcc

Driver CAN

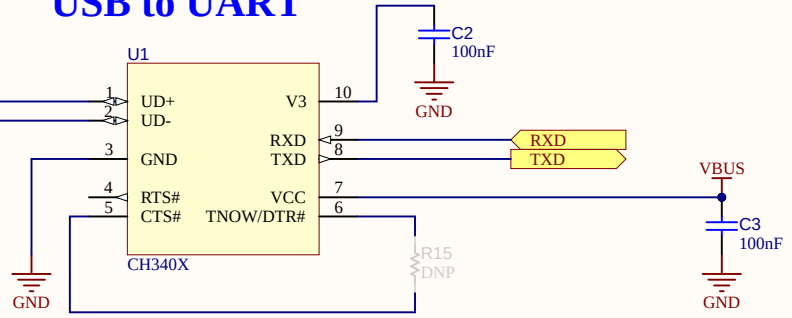


CC voltage (typ.)	Host advertises	Meaning
~0.2–0.4 V	Default USB current	500 mA (USB 2.0) / 900 mA (USB 3.x)
~0.9–1.2 V	1.5 A@5V	High-current USB-C
~1.6–2.0 V	3.0 A@5V	Maximum without USB-PD

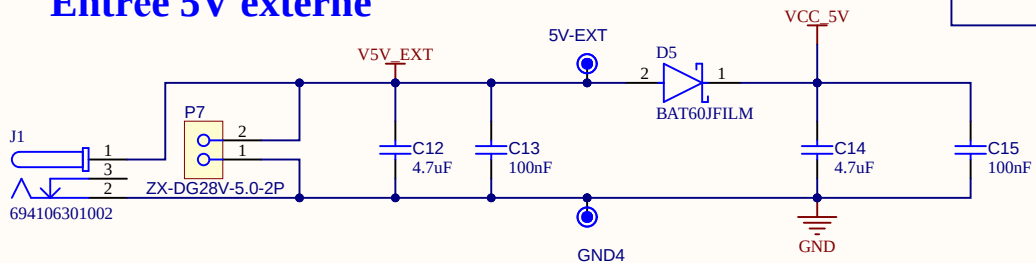
Interface USB et CAN



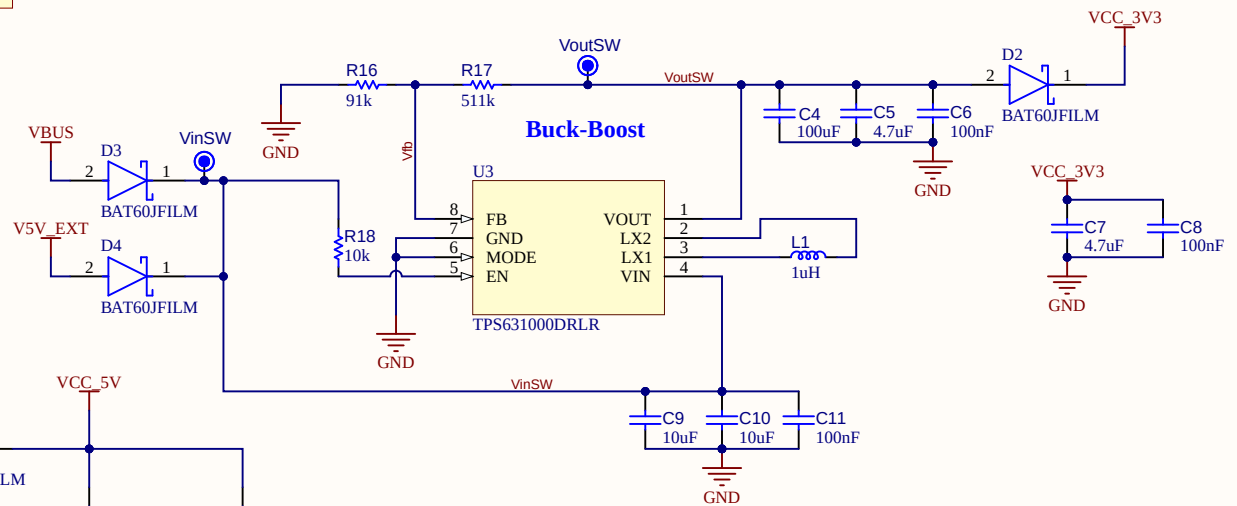
USB to UART



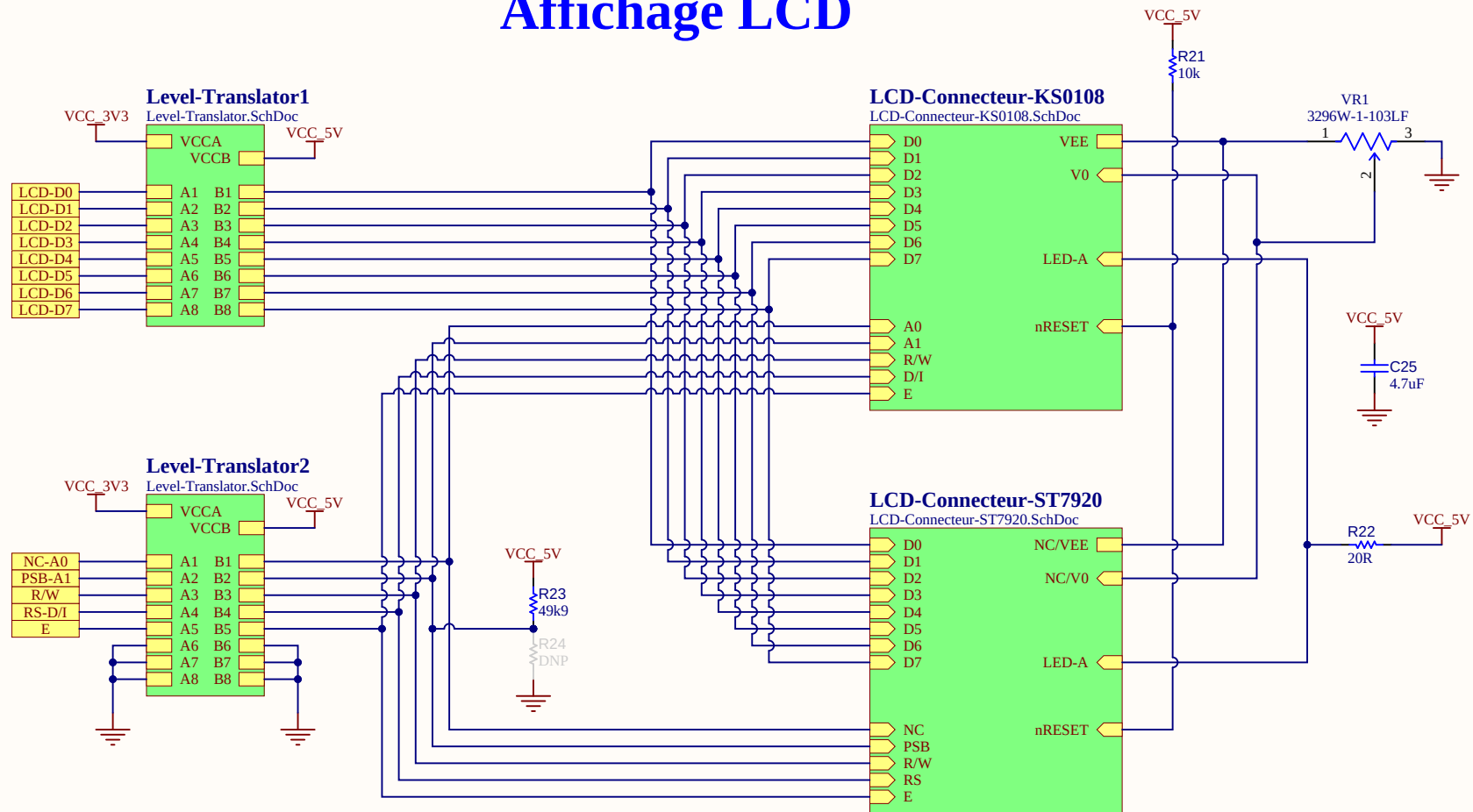
Entrée 5V externe



Buck-Boost



Affichage LCD



Titre : Affichage LCD

Conception :
CEGEP Limoilou

Date : 2026-02-27

Fichier : Affichage-LCD.SchDoc

Approbation :
kc

Heure : 13:54:47

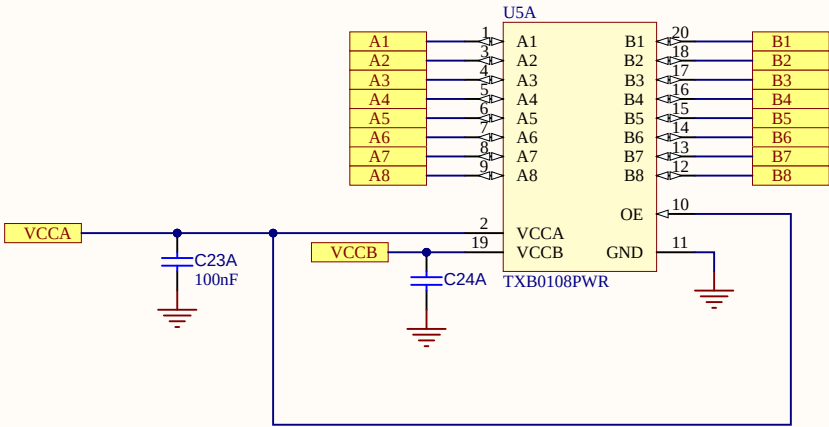
Révision : Rev.B

Feuille 7 de 10

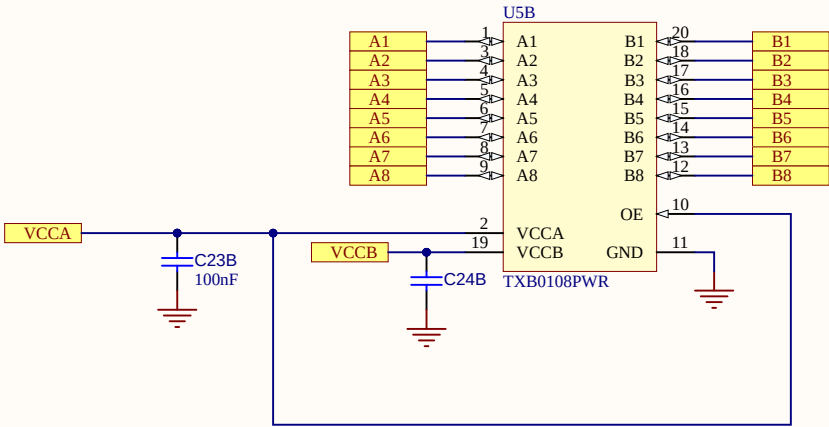
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Bidir Level Translator

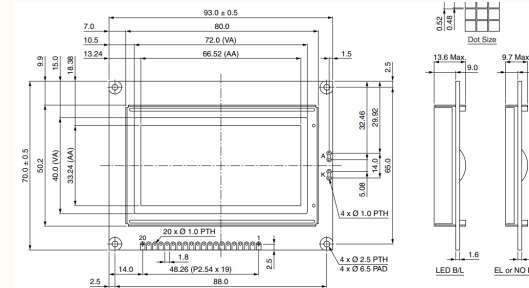
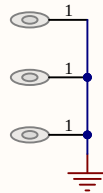
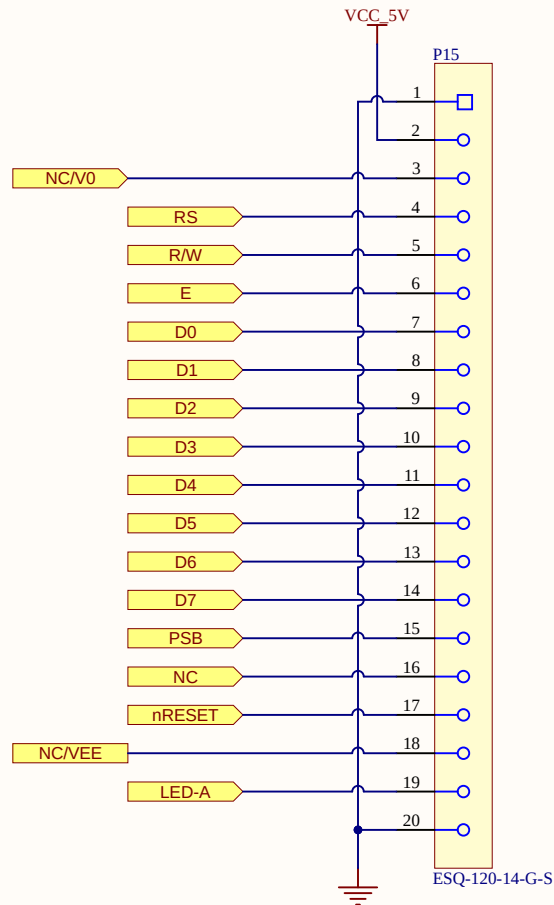


Bidir Level Translator



LCD - Connecteurs

Option LCD 12864 70x93mm



Pin No.	Symbol	Function
1	V _{SS}	Ground
2	V _{DD}	Power supply (+5.0V)
3	V ₀	Contrast adjustment
4	R _S	Data/Instruction
5	R _W	Data read/write
6	E	H → L enable signal
7	DB0	Data bus line
8	DB1	Data bus line
9	DB2	Data bus line
10	DB3	Data bus line
11	DB4	Data bus line
12	DB5	Data bus line
13	DB6	Data bus line
14	DB7	Data bus line
15	CS1	Chip select for IC1
16	CS2	Chip select for IC2
17	/RES	Reset
18	V _{EE}	Negative voltage output
19	A	Power supply for LED (+5.0V, I _{LED} = 0.0)
20	K	Power supply for LED (0V)

Pin Description and Wiring Diagram

Pin No.	Symbol	External Connection	Function Description
1	V _{SS}	Power Supply	Ground
2	V _{DD}	Power Supply	Supply Voltage for Logic (+5.0V)
3	V ₀	Adj. Power Supply	Supply Voltage for Contrast (approx. -3.7V)
4	R _S	MPU	Register Select: 1=Data, 0=Instruction
5	R _W	MPU	Read/Write select signal, R/W=1: Read R/W=0: Write
6	E	MPU	Operation Enable signal. Falling edge triggered.
7-14	DB0-DB7	MPU	This is an 8-bit Bi-directional data bus
15	CS1B	MPU	Chip Selection: CS1=H, CS2=L → select IC1 (left side)
16	CS2B	MPU	CS1=L, CS2=H → select IC2 (right side)
17	/RES	MPU	Active LOW Reset signal
18	VEE	Power Supply	Negative voltage output (-10V)
19	LED+	Power Supply	Backlight Anode (+5V Via On Board Resistor)
20	LED-	Power Supply	Backlight Cathode

Titre : LCD - Connecteurs

Conception :
CEGEP Limoilou

Date : 2026-02-27

Approbation :
kc

Heure : 13:54:47

Révision : Rev.B

Feuille 9 de 10

Fichier : LCD-Connecteur-ST7920.SchDoc

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Québec, QC
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LCD - Connecteurs

Option LCD KS0108
50x75mm

